

1. How does Power BI handle large datasets in the Online Service, and what is the role of Premium Capacity in this?

Power BI handles large datasets using **import compression techniques**, **incremental refresh**, and **data modeling best practices**.

Premium Capacity enables datasets up to **400 GB** (vs 1 GB in Pro), offers **dedicated resources**, **higher refresh rates**, and **enhanced performance**—making it ideal for enterprise-scale data models.

2. What are the differences between Import mode, DirectQuery, and Live Connection in Power BI Service?

Mode	Description	Performance	Data Refresh
Import	Data is loaded and stored in Power BI.	Fast	Needs scheduled/auto refresh
DirectQuery	Queries run directly on the data source (no import).	Slower	Live; no storage in Power BI
Live Connection	Connects to SSAS or Power BI dataset (semantic model).	Fast (query SSAS)	Real-time, no dataset in workspace

3. Explain deployment pipelines in Power BI Online. What stages do they include?

Deployment pipelines streamline **report lifecycle management** across environments. They include **three stages**:

- **Development:** Build and test reports and datasets.
- **Test:** Validate with controlled user groups.
- **Production:** Final live version for business use.

Each stage can have its own workspace. Changes can be **deployed forward** or **rolled back**.

4. How can Power BI Service integrate with Microsoft Teams or SharePoint for collaboration?

- In **Teams**: Add Power BI reports as a **tab**, use **Power BI app for Teams**, or chat about visuals directly.

- In **SharePoint**: Embed reports using “**Power BI web part**”, allowing secure and interactive viewing.
This improves **team access** and **real-time discussion** around data.
-

5. What is the XMLA endpoint in Premium and how does it benefit developers or enterprise BI teams?

XMLA Endpoint allows **read/write access** to Power BI datasets using **SSMS, Tabular Editor, and Excel**.

Benefits:

- Advanced data modeling
 - Automation through DevOps
 - Integration with enterprise tools
Only available with **Power BI Premium/Premium Per User (PPU)**.
-

6. Describe how usage metrics and audit logs work in Power BI Service.

- **Usage metrics** (built-in reports): Show how often a report is viewed, by whom, and when.
 - **Audit logs** (via Microsoft 365 Compliance Center): Track user activity like sharing, viewing, exporting, publishing.
These tools help monitor adoption, detect unusual activity, and support compliance.
-

7. How do you manage workspace access and permissions for different users?

Power BI workspaces use **role-based access control**:

- **Admin**: Full control.
- **Member**: Edit, publish, and delete content.
- **Contributor**: Publish and edit content.
- **Viewer**: View-only access.

Permissions can be adjusted per **dataset**, **report**, or **app** for fine-grained control.

8. How can data governance be enforced in Power BI Service?

- **Sensitivity labels** (Microsoft Purview)

- **Row-Level Security (RLS)**
- **Certified/Promoted datasets**
- **Approval workflows**
- **Audit logs & activity monitoring**

Together, they enforce **data protection, quality, and compliance policies** within the organization.

9. What are the limitations of Row-Level Security when using DirectQuery or Live Connection?

- **Performance impact:** Filtering at source can be slow with large data volumes.
 - **Limited by source:** Some sources (e.g., SSAS Tabular) manage security outside Power BI.
 - **RLS not supported in Live connections** to Power BI datasets (must be defined in source).
 - Can be bypassed if users have **direct access to source system**.
-

10. Explain how you can refresh a dataset via Power Automate or REST API.

- **Power Automate:** Use the "**Refresh a dataset**" action from Power BI connector to trigger refresh based on events (e.g., form submission, new data).
- **REST API:** Developers can use **POST** https://api.powerbi.com/v1.0/myorg/datasets/{dataset_id}/refreshes to trigger programmatic refreshes (suitable for scheduled jobs, DevOps).